

Fast, Reliable, Correct

Module 11 – monitoring and evaluation

Get to know Lana — Fun Fact



We're going to Europe today

- Working in a time zone only one hour from you!!!!
- Living and working in Spain for July and August
- We hope to move to Spain next year.

Housekeeping

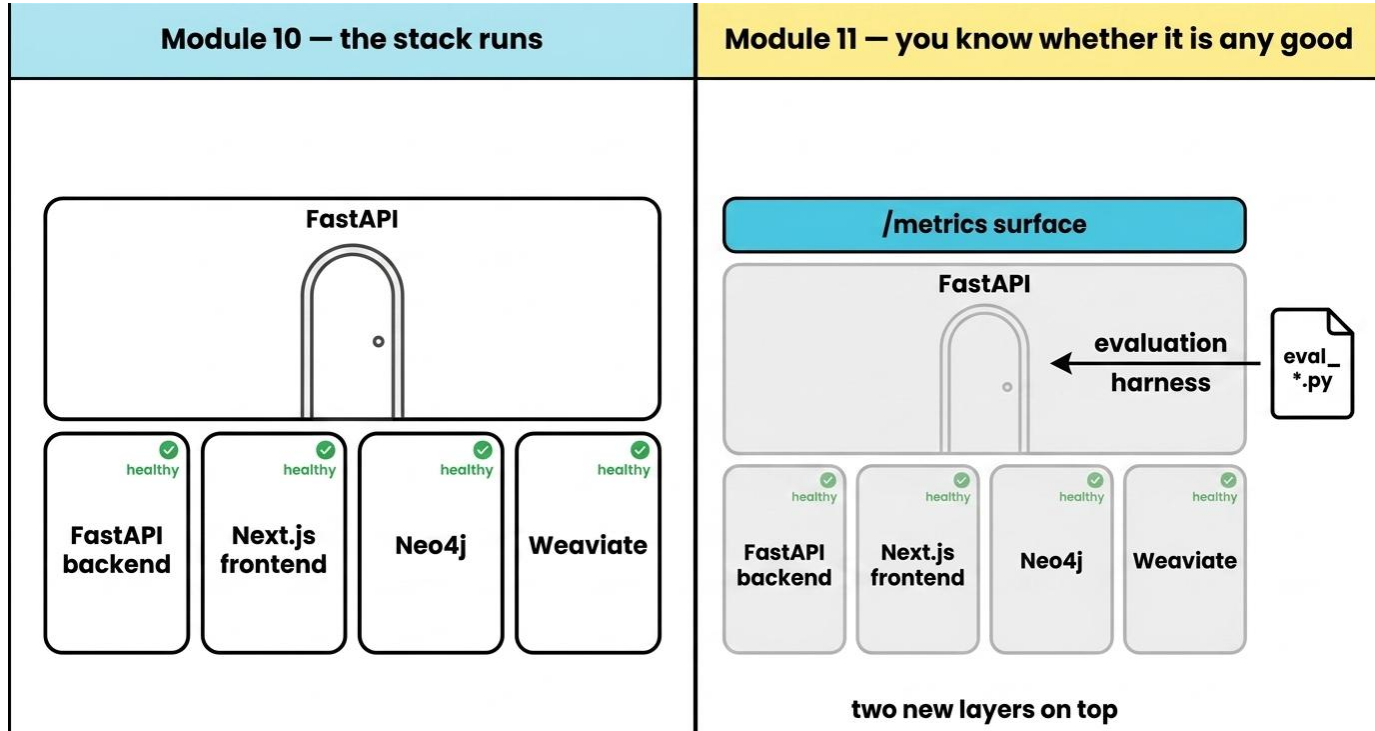
“Cleaning up” as we Enter the Last Module

“Housekeeping”

Resubmissions will be allowed 1 week after
Capstone

Your LinkedIn needs to be “student” not
intern of LevelUp Economy

From Module 10 To Module 11



Who is the French president?

Bertrand Russell :

Service A	Service B
p95 latency 50 ms	p95 latency 800 ms
5xx error rate 0%	5xx error rate 2%
<i>"The king of France is bald."</i>	<i>"France is a republic with a president, not a king."</i>

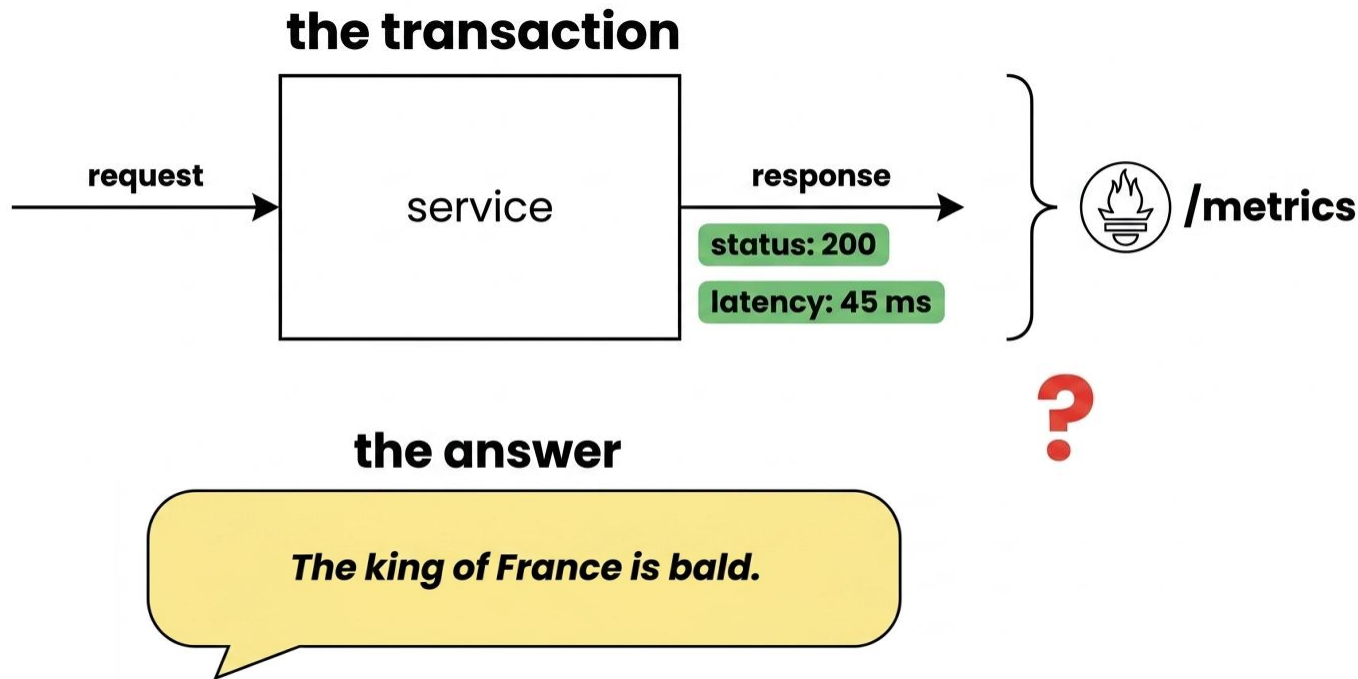
Same question. Two different services. Both respond.

Monitoring

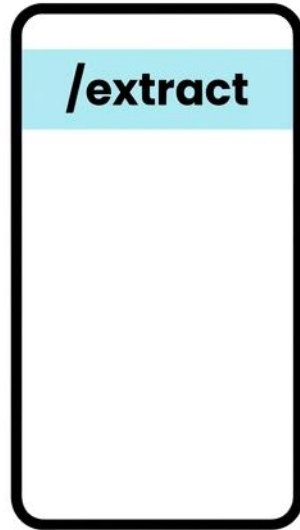
A Service Dashboard: Fast And Reliable Are Not Enough



Why Quality Is Invisible in Prometheus



Metrics As A Contract



client-facing API

```
class Req(BaseModel):  
    text: str  
  
class Resp(BaseModel):  
    ==  
    ==
```



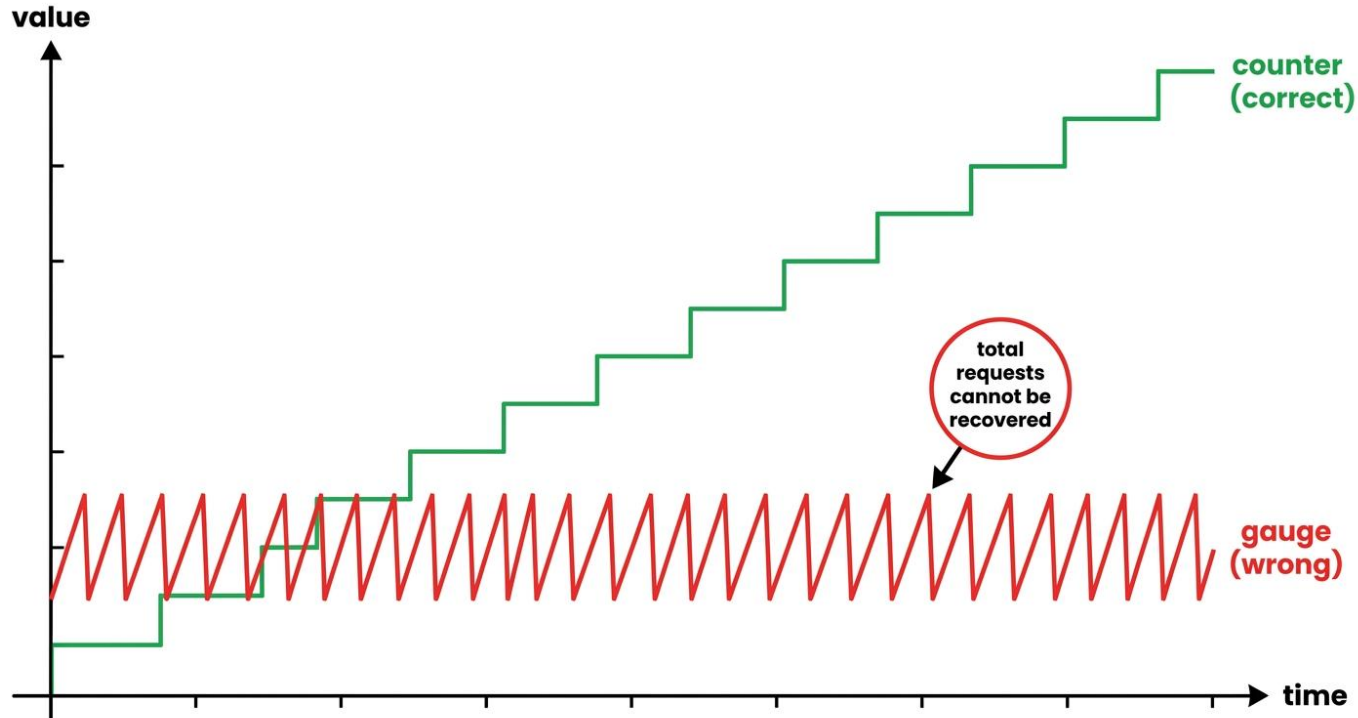
dashboard + script-facing API

```
# HELP requests_total Total HTTP requests  
# TYPE requests_total counter  
requests_total 1247
```

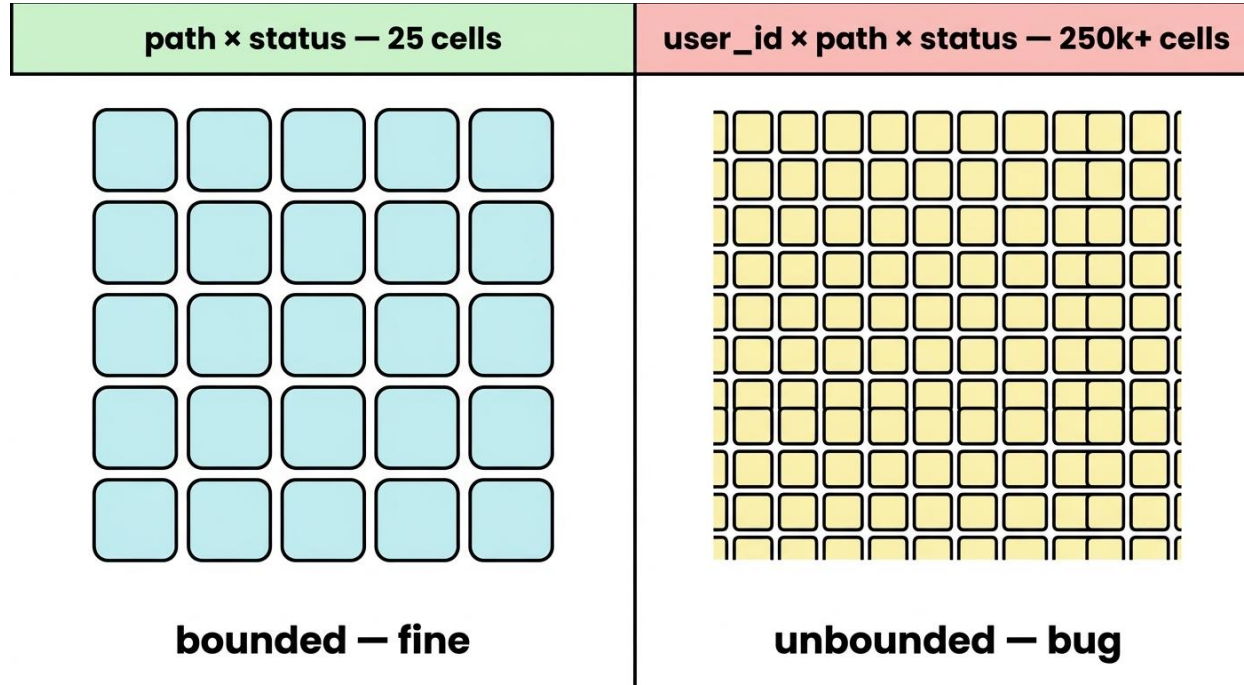
The Four Metric Types

counter	gauge	histogram	summary
			
only goes up	up or down	bucketed distribution	client-side quantiles
total requests	in-flight requests	request latency	

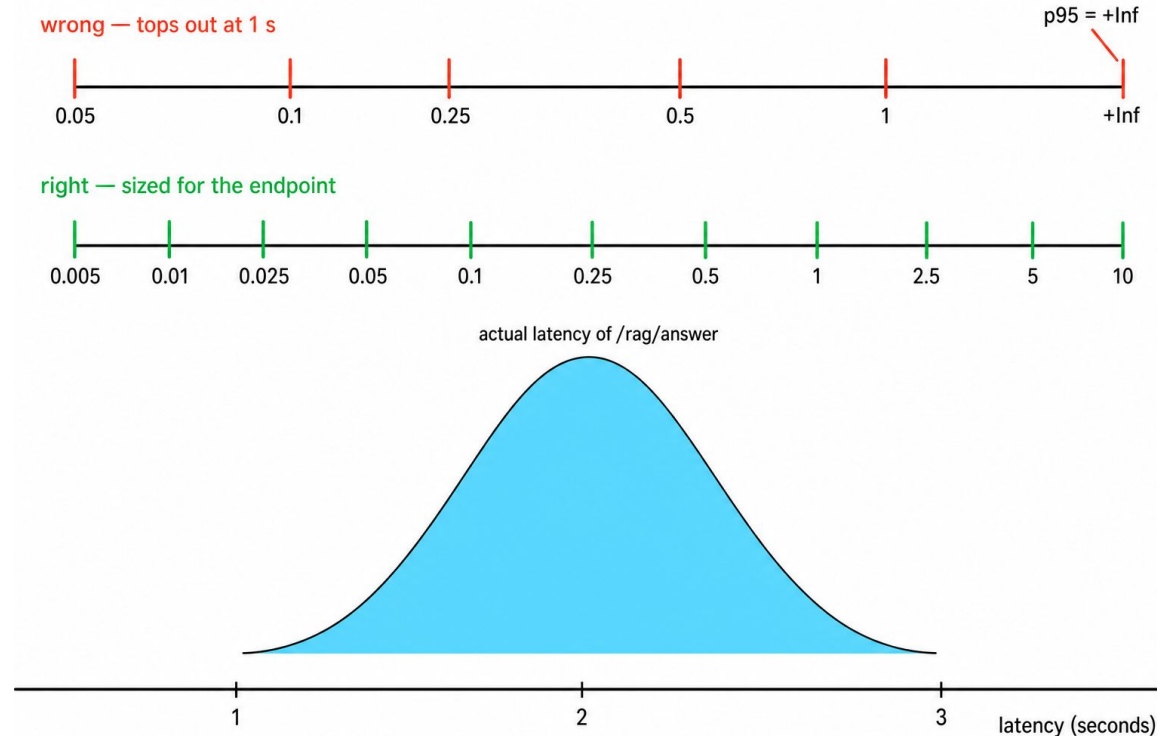
Wrong Type, Wrong Truth



Label Cardinality



Histogram Buckets



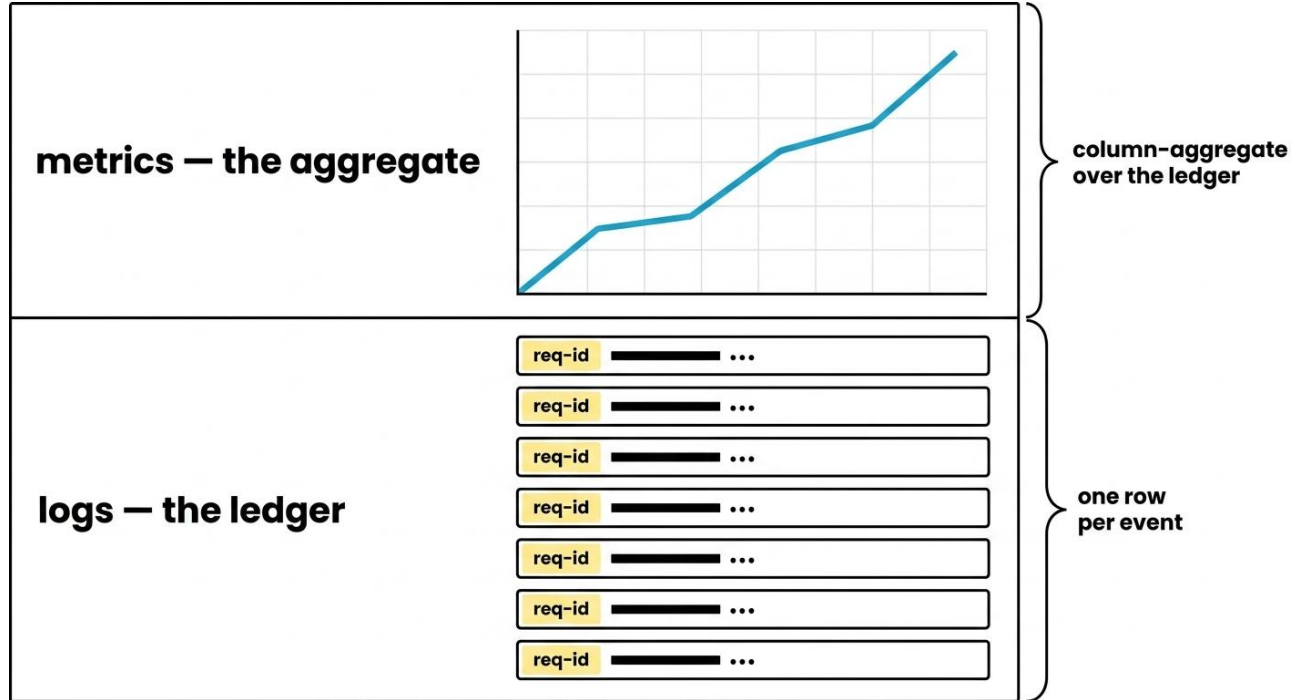
A Quick Look At `/metrics`

```
$ curl http://localhost:8000/metrics

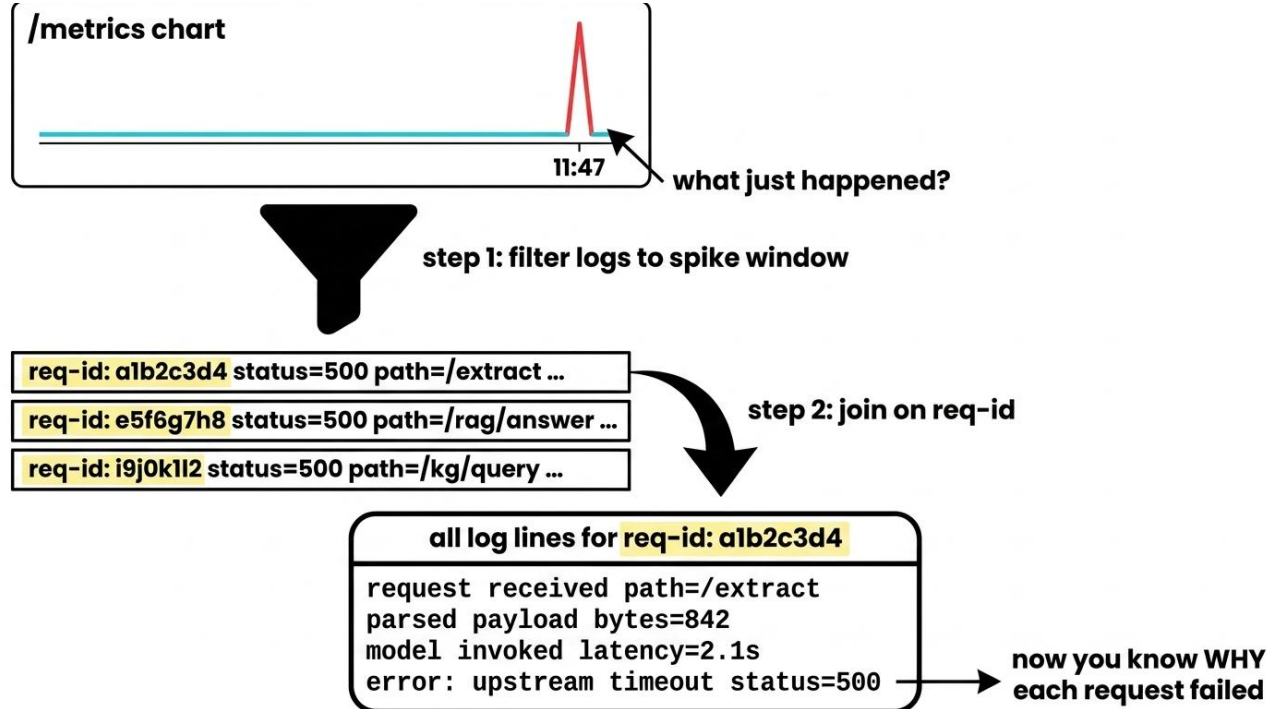
# HELP http_requests_total Total HTTP requests
# TYPE http_requests_total counter
http_requests_total{path="/extract",status="200"} 1247
http_requests_total{path="/extract",status="500"} 3
http_requests_total{path="/kg/query",status="200"} 612
http_requests_total{path="/kg/query",status="500"} 4
http_requests_total{path="/rag/answer",status="200"} 287
http_requests_total{path="/rag/answer",status="500"} 2

# HELP request_latency_seconds Latency
# TYPE request_latency_seconds histogram
request_latency_seconds_bucket{le="0.05"} 980
request_latency_seconds_bucket{le="0.1"} 1050
request_latency_seconds_bucket{le="0.25"} 1120
request_latency_seconds_bucket{le="0.5"} 1170
request_latency_seconds_bucket{le="1.0"} 1195
request_latency_seconds_bucket{le="2.5"} 1230
request_latency_seconds_bucket{le="5.0"} 1248
request_latency_seconds_bucket{le="+Inf"} 1250
request_latency_seconds_sum 142.7
request_latency_seconds_count 1250
```

Two Layers Of Observability



Request-ID Is The Join Key

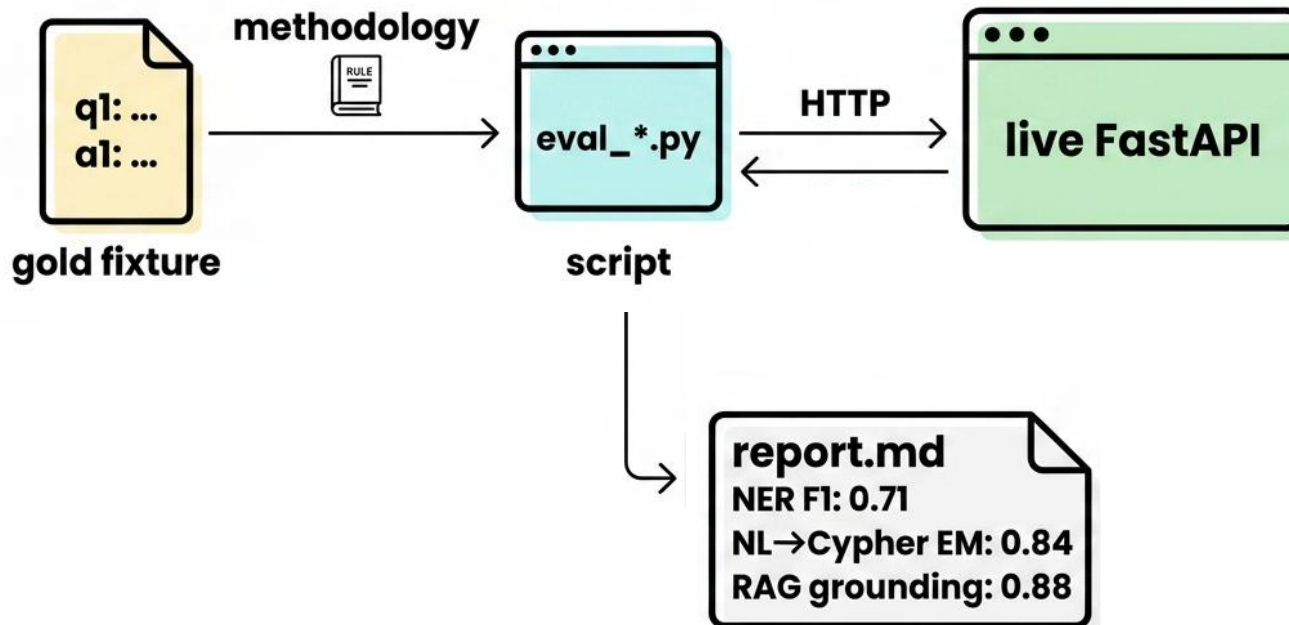


Logs Are Data

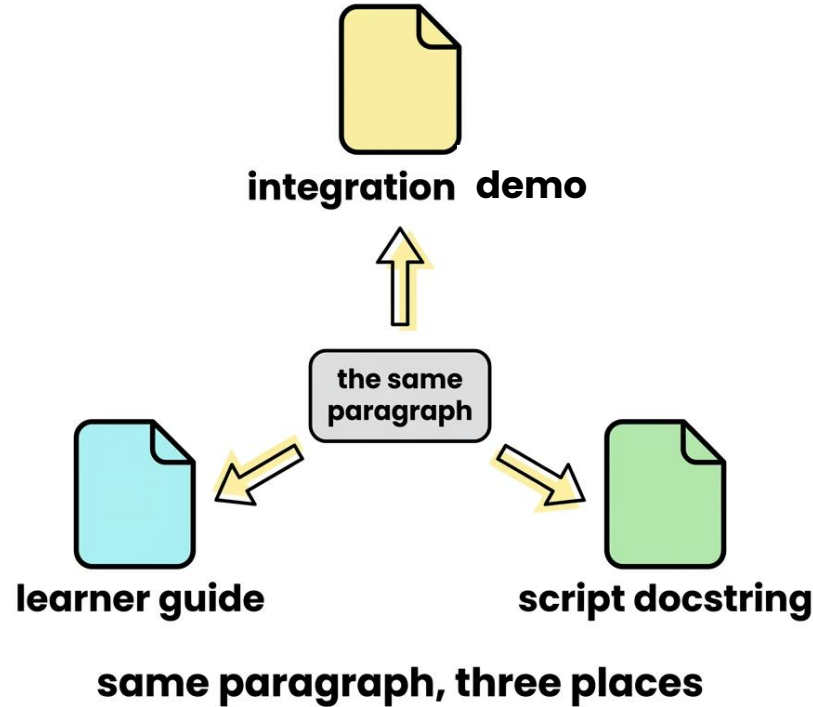
print()	structured JSON
"request done in 45ms"	parseable by every tool you will ever own <pre>{ "timestamp": "2025-04-12T11:47:02", "level": "info", "request-id": "a1b2c3d4", "path": "/extract", "status": 200, "latency in milliseconds": 45, "free-form message": "request done" }</pre>
"completed: status=200"	<pre>{ "timestamp": "2025-04-12T11:47:03", "level": "info", "request-id": "e5f6g7h8", "path": "/extract", "status": 200, "latency in milliseconds": 62, "free-form message": "completed" }</pre>
"error processing user request"	<pre>{ "timestamp": "2025-04-12T11:47:04", "level": "error", "request-id": "i9j0k1l2", "path": "/extract", "status": 500, "latency in milliseconds": 120, "free-form message": "error processing user request" }</pre>

Evaluation

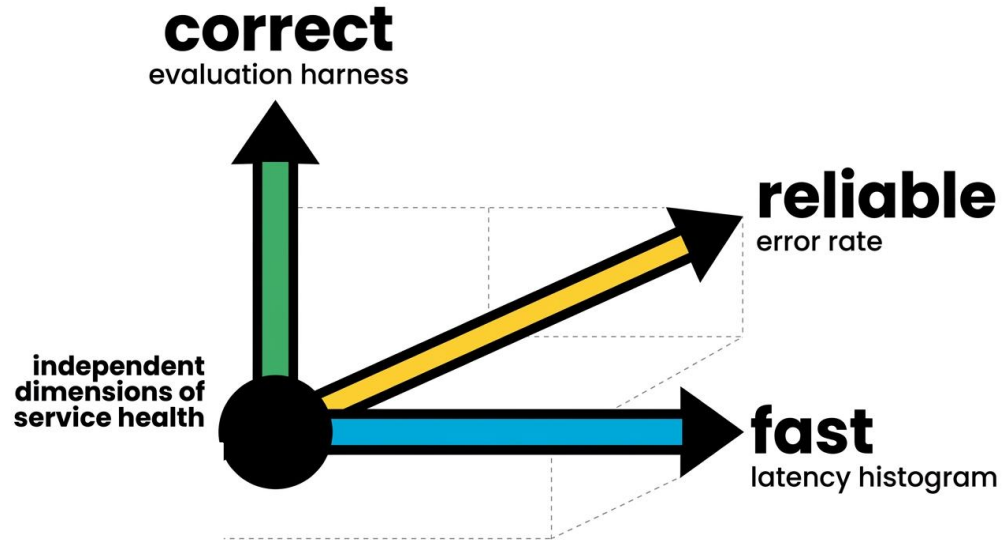
Quality Is Its Own Discipline



Methodology Restated



Service Health



Service A vs. Service B

neither is fully healthy

Service A
p95 latency / 50 ms
5xx error rate / 0%
<i>"The king of France is bald."</i>

- fast
- reliable
- correct

Service B
p95 latency / 800 ms
5xx error rate / 2%
<i>"France is a republic with a president, not a king."</i>

- fast
- reliable
- correct

Questions